

Phase 2 Report

Group 25

Matthew Pham : 301470163

Kaitlyn Lum : 301558008

Ryan Fung : 301579052

Marika Screpnek : 301560632

Faculty of Applied Science, Simon Fraser University

CMPT 276: Introduction to Software Engineering

Dr. Saba Alimadadi

Hossein Soltanloo

Nima Modares Gorji

CMPT 276 D100

March 11, 2025

Approach:

Our approach to implementing phase 2 of the project was to partition tasks between members and share several tutorials to ensure our different portions were compatible. None of us were too familiar with Java, therefore we used online resources to code and learn as we went including videos and articles. We had meetings every Monday and Thursday, which we used to share personal timelines and plans, and communicated frequently through Discord.

We began with simple aspects of the game such as the board and main character. Then we implemented the moving enemy followed by coins, lost notes, cones, and parked cars. We then implemented a score and a timer. Finally, we created screens and implemented their functionality depending on various situations (win/loss/instructions/main menu).

Modifications to Initial Design:

The main modifications we carried out during phase 2 were changes to our UML class diagram. `removeEntity()` was not needed in the Entity class as the Board class should be responsible for the removal of entities and their attributes; we created several methods to remove the Entities from the Board. We also made the Entity class abstract because it provided a blueprint for classes such as moving and static entities, and we never needed to create an instance of the Entity class.

Methods were added to get the image from the resources and draw the image onto the board. We added a constructor to the MainCharacter class as the location of the image needed to be fixed when shown on the board. One major change we had was removing and implementing the time class into the board class. This made the game state and tracking the time easier because getter methods were accessible in the board class and its subclasses. For the Reward class, we created only a constructor and removed the collected boolean and `isCollected()` since the Cell class was not needed and was not created; the collected attribute

was meant to check if the Cell contained a Reward. Instead, we checked if the integer X and Y positions of entities and if any matched, then that would be considered a collision. Since we did not need a Cell class, we did not need to have setEntity (setCollected) and checkEntity (isCollected). In contrast, we created a collectCoins() method in Board class to allow the main character's position to be checked if it is the same as a coin's position; similarly, this also applies to collectLostNote(). For these collect methods, we created an ArrayList to store their unique types. We created a method generateCoords() in the LostNote class to help generate a random location within the dimensions of the board, to be populated on the board. We created methods in Board class to read the images of the Entities. We added a "main menu" button to the end screen to allow the user to choose whether they want to play again or exit out of the game. We created an abstract LoseScreen class with subclasses LoseScreenConcord and LoseScreenScore. This way there were no if statements in the LoseScreen class, and multiple instances of the class wouldn't be created when not necessary.

Management Process:

Our management process stayed mostly consistent with what was highlighted in phase 1 of our project. We began with a scheduling meeting, deciding which elements we wanted to implement by the halfway deadline. We decided on the main board, main character, and moving enemy. Matthew and Marika implemented the main board together because in planning we figured that it would be the most intensive task of the 3. They did this by messaging on discord and staying in communication while coding. Kaitlyn implemented the main character and Entity class, and Ryan implemented the moving enemy.

Following the halfway deadline, we discussed the remaining tasks and partitioned them between members. The remaining tasks were score, timer, coins, lost notes, cones, parked cars, and screens such as the main menu, instruction, win, and lose. Matthew did the

score and timer, Kaitlyn did the coins and lost notes, Ryan did the cones and parked cars, and Marika did the screens. Finally, we met on the Monday before the final deadline to write the report together as a group and finished the initial tasks on Monday and Tuesday.

External Libraries:

Java.swing - We used this Java GUI toolkit that was demonstrated in several online tutorials. We decided to use this library for GUI because there were many accessible online resources to learn the capabilities. We used this library for creating windows, different panels, and buttons.

Java.awt - We used Java AWT for simpler GUI components such as painting components, and updating game states with repaint. We also used AWT for action/event handlers.

Javax.imageio - We used Java Image I/O to retrieve images from the resources folder and assign them to their respective entities.

We used a combination of these libraries for the GUI based on which libraries were typically used in tutorials for different functionality.

Code Quality:

To maintain code readability each member commented on their code as they wrote it. Additionally, members who wrote a certain class implemented JavaDocs describing the purpose and functionality of the class, their fields, and their methods. Unused methods and attributes were deleted from their corresponding class. We tried to use methods to help make the implementation more readable.

Challenges:

During Phase 2, some challenges we faced were merge conflicts, modifications to the UML class diagrams, and conflicts with personal schedules and the project schedule. During

the changes to the UML diagrams from Phase 1, we had to decide what methods or attributes needed to be removed or added. Also, time management was a challenge with personal commitments, exams and other assignments affecting the progression of our project.